

Institute of Actuaries of India

Subject ST4 — Pensions & Other Employee Benefits

May 2012 EXAMINATION

INDICATIVE SOLUTIONS

Solution 1 :**(i)**

The political, economic and fiscal viewpoints of the State will determine the precise roles that it will play. However, the roles are likely to fall within the following categories :

- provide benefits to some or all of the population
 - educate or require education about the importance of providing for the future
 - regulate to encourage or compel benefit provision by or on behalf of some of the population
 - regulate bodies providing benefits, and those bodies with custody of funds, in an attempt to ensure security for promises made or expectations created.
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- provide benefits to some or all of the population – Civil servant pension system or National old age pension scheme
 - educate or require education about the importance of providing for the future – National pension scheme (NPS)
 - regulate to encourage or compel benefit provision by or on behalf of some of the population – Provident fund, Gratuity, Superannuation, Employee deposit linked insurance, Employee state insurance
 - regulate bodies providing benefits, and those bodies with custody of funds, in an attempt to ensure security for promises made or expectations created – Insurance, mutual funds, banking products, post office offerings

(ii)

The commercial objectives of employers in providing benefits are as below:

- controlling costs
- providing benefits that employees perceive to be attractive
- providing benefits that are at least in line with those offered by competing employers
- taking advantage of any financial incentives from the State, e.g. tax incentives
- rewarding certain classes of employees, e.g. loyal staff.

[5]**Solution 2 :**

Risks facing the sponsor of a Defined Benefit pension scheme are as follows:

- Investment risk – Return on existing investments may turn out to be lower than expected leading to additional contributions to be made by the sponsor
- Reinvestment risk – Risk that future contributions and investment proceeds after will need to be invested under unfavourable investment conditions leading to a lower yield on amounts invested in future
- Default risk – Risk that the issuer of investment instrument will default or become insolvent leading to a capital loss
- Asset-liability mismatch risk – Risk that value of liabilities will increase more than the value of assets under certain macro-economic scenarios
- Opportunity cost – There is a risk that the return on scheme assets turn out to be lower than the returns that could have been achieved had those funds been invested internally within the company
- Longevity – Post-retirement mortality experience may turn out to be lighter than expected, leading to overall increase in scheme costs

- In-service mortality could be different from expected
- Salary increments may turn out to be higher than expected
- Scheme withdrawal experience could be different from expected
- Higher than expected benefit improvements
- There is a risk of mis-estimating the amount and incidence of contributions due to:
 - The use of an unsuitable model
 - The use of unsuitable parameters
 - Errors in membership data or data used to derive parameters
- Contributions may be required at inopportune times at amounts higher than expected, depending on the overall scheme experience
- Risk of inadequate funds or under-funding to pay the benefits as they fall due, leading to higher than expected contributions being paid to the fund, at a time when the sponsor has insufficient liquid assets to pay these additional contributions
- Risk of excess funds leading to:
 - A risk that sponsor may not be able to use these excess funds for other purposes
 - A risk of financial penalty being imposed by tax authorities who may view this as tax-evasion
- Illiquidity risk – Even if the scheme is adequately funded, there is a risk that the sponsor may not be able to liquidate the assets to pay out the benefits as they fall due.

Risks related to benefits

- Benefit changes – There may be a change in regulation leading to a higher benefits being paid to scheme members, leading to overall increase in cost
- Risk of unappreciated benefits – If scheme benefits are not appreciated by members/beneficiaries, it can be thought that the capital used to provide these benefits has not been put to the best use

Risks related to general operation of the scheme

- Fraud or misappropriation by an employee or a third party
- Inappropriate advice
- Incorrect benefit payments
- Higher **than expected administrative costs**
- **Penalty or fines by regulatory authorities**

Risks related to regulation and taxation

- Change in regulatory environment leading to increased contributions
- Changes in taxation rules

[5]

Solution 3 :

(i)

Sponsor's perspective:

- Book reserving provides greater flexibility for the sponsor compared to financing methods where explicit funds are set aside, since the risk of making contributions at inopportune times is substantially reduced
- No specific assets are earmarked for pension scheme liabilities, therefore contributions remain totally self-invested and are available for normal business activities of the scheme sponsor

Member's perspective: The main problem with book reserving is the lack of security compared to other financing methods

- In the event of insolvency of the sponsor, the members may not receive full benefits
- The benefits received by the member will depend on:
 - Whether the sponsor had taken any insolvency insurance
 - Winding-up priority of book reserves

- Whether any third-party guarantee is available
- However, book reserving would imply more capital would be available for normal business activities, which might result into greater benefits for the members in the long run

(ii)

The factors that the sponsor would need to consider are as follows:

1. Opportunity cost

- Since the company is rapidly growing, it is likely that it would require capital to support normal business activities. From a financial perspective, the company should set aside funds only if it would get a higher effective investment return compared to what it would get by investing the funds within the business
- Setting aside funds will substantially increase the security of promised benefits in eyes of the scheme members

2. Flexibility

- Financing the benefits would require that regular contributions be made at regular intervals. Call for such contributions may be made at inopportune time. Company would ideally want to have flexibility with regard to payment of contributions
- Any deficit within the fund needs to be made up by the company while a surplus may not be reclaimed by the employer

3. Regulation

- Company would need to consider any regulatory issues with regard to financing of benefits

4. Taxation

- Company would need to consider any tax issues involved. Tax benefits might be available for the company depending on legislation which could enhance the effective net of tax return on scheme assets

(iii)

The company could go for any of the following methods of “funding” the benefits:

1. Lump-sum in advance

- Under this method, a lump-sum contribution is made when a member joins the scheme
- Entire funding is done even before the first benefit payment is made
- The method would tie up excessive capital within the scheme which could have been put to a better use within the business
- Excessive fund build up within the scheme might be viewed unfavorably by tax authorities, who might impose a penalty for over-funding
- The method would provide the maximum security for the promised benefits as the contribution is expected to cover all scheme benefits

2. Terminal funding

- Under this method, a contribution is paid whenever a benefit starts to be paid
- The contribution is designed to cover all future benefit payments
- The method does not provide any security of past service (or accrued) benefits
- Terminal funding would require payment of huge sums of money within the scheme to secure future benefits. This might result into liquidity problems
- **Regular contributions**
- Under this method, contributions are made at regular intervals to meet promised scheme benefits
- Greater flexibility than other two advance funding methods with regard to timing and stability of contributions
- Greater security than terminal funding as a fund gradually builds up over the service period of the member, rather than at the end of service period

- A host of funding methods available to meet company's desired funding criteria – e.g. security, stability, durability, realism etc.

[13]

Solution 4 :

The possible sources of loss would fall in three areas

- Assumptions gain/ loss
- Experience loss
- Enhancement in plan benefits
 - The assumptions gain/ loss would arise due to change in assumptions during the year from 31 March 2011 to 31 March 2012
 - For example a decrease in discount rate or increase in medical inflation basis will give rise to an assumption loss
 - The experience loss could be due to the following reasons
 - The probability of providing the benefit
 - lower attrition than assumed leading to higher number of members becoming eligible for the benefit
 - lower deaths (better longevity) than assumed
 - The expected cost of the benefit
 - Higher increase in average hospitalisation cost than assumed
 - Higher benefit payments than expected during the year

[5]

Solution 5 :

(i)

The following factors need to be considered while setting up a DB scheme

- What level of benefit should be provided?
- Should the benefits be defined in monetary or real terms?
- If in real terms should they be related to the purchasing power or earnings?
- On what events should benefits be provided, e.g. on death, on retirement, on leaving employment, whilst unable to work, during employment, on divorce?
- In what form should the benefits be provided, e.g. an annuity, a cash lump sum, goods, services?
- Should the approach differ between individuals in the group?
- If they should differ, in what way, e.g. form, level, real nature, events?
- Should the individuals be given any options about the events, forms or levels of benefits?

(ii)

Background:

We understand that the Company currently provides the Gratuity benefit that is better than the Act. The company would like to explore options to understand options for redesign of the benefit so that the costs for providing the benefits are optimised.

Objective:

The Company would like to optimise the spend by redesigning the benefit.

Options for redesign:

- The options for redesign would depend on the inputs provided by the Company on the aspects to be considered for redesign
- Some of the possible options could be

- Grandfather the benefit and provide benefit as per Act to new entrants
- Provide option to the existing employees to cash out the additional future accrual and provide benefit as per Act to those who opted to cash out and the New entrants

Aspects to be considered for redesign:

- What level of benefit should be provided? While the benefit cannot be less than the mandatory benefit, the company could choose to provide a higher level of benefit than the mandatory minimum
- Should the benefits be defined in monetary or real terms? Would the company want to limit the benefit at 10 lakhs as per act or a multiple of salary or any other limit
- Does the Company want to provide a death in service insurance benefit if not being provided now
- Should the approach differ between individuals/ grades/ age groups/ service groups in the Company
- If they should differ, in what way, e.g. level (accrual rate), real nature (multiple of salary), events (vesting criteria)?
- Should the individuals/ grades/ age groups/ service groups be given any options about the levels of benefits? The company can provide flexibility to employees to cash out the future additional benefit for instance.

[10]

Solution 6 :

(i)

- Sponsor may be unwilling to finance future benefits under the scheme, because:
 - Existing members are being transferred into a new scheme for future benefit accrual
 - The sponsor may be experiencing financial difficulties because of which it is unable to continue financing the scheme benefits
- The trustees may have decided to discontinue the scheme if they think this would be the best way to secure “accrued” benefits because of a weak sponsor’s covenant

(ii)

The discontinuance benefits would depend on the following factors:

1. Rights of beneficiaries

- Rights may be viewed as what the individual would receive if they:
 - left the scheme voluntarily at the moment of the discontinuance
 - remained in the scheme until retirement (but without any further accrual)
 - remained in the scheme until retirement and continued to accrue benefits
- Trustees would need to consider the terms of the scheme to understand the discontinuance benefits explicitly mentioned in case of scheme discontinuance
- Trustees would also need to ensure that they comply with any overriding legislation

2. Expectations of beneficiaries

- Members would expect their past service benefits to increase in line with salary growth
- If there had been any practice within the scheme of awarding discretionary benefits, such as pension increases or enhanced early retirement terms, there may be an expectation that these will continue
- Contributions paid by or on behalf of the beneficiaries may also create an expectation that the benefits will be at least as valuable as these contributions
- The existence of any minimum funding target, whether by statute or voluntary practice of the scheme, may also give the beneficiaries the expectation that the funding target will be achieved and that they will receive the benefits to which the target relates

3. The level of assets (maximum 3 marks)

- Rights and expectations of members are only relevant if sufficient assets are available
- If the assets are insufficient to meet the rights of the beneficiaries then a lower benefit will be paid
- Where the benefits have to be reduced, legislation or scheme rules may indicate which types of benefits are to be reduced or which types of beneficiaries are to have their benefits reduced
- Expenses related to discontinuance process may need to be paid in full before any benefits could be paid out from the fund
- Consideration needs to be given to any additional funds that might become available from the sponsor on account of legislation or insolvency insurance
- If the assets are more than the liabilities, consideration must be given to legislation, scheme rules, sources of surplus and rights and expectations of members before deciding the amount of surplus that could be refunded to the sponsor

4. Method of provisioning

- Method of provisioning for discontinuance benefits will also have an impact on level of benefits made available to the members
- If liabilities are liquidated or transferred to a third party, there would be a cost involved in disinvesting and transferring assets. These costs have to be borne out by the scheme in full before discontinuance benefits could be decided
- The amount of benefits could further depend on the benefit options available with the third party and these may be different from scheme'

(iii)

The following methods are available for the trustees to provision for the outstanding benefit payments:

1. Gradual removal of liabilities by the continuation of the scheme without any further accrual of benefits.
2. Transfer of the liabilities to another pension scheme with the same sponsor
3. Transfer of the funds to the beneficiary to extinguish the liability
4. Transfer of the funds to an insurance company to invest and provide a benefit
5. Transfer of the liabilities to an insurance company to guarantee the benefits
6. Transfer of the liabilities to a central discontinuance fund, operated on a national or perhaps industry-wide basis

[13]**Solution 7 :****(i)**

The design features to be considered in setting up a defined contribution scheme are

- The level of benefits
- The affordability of contributions
- Market practice and alignment to market
- Segmented approach based on age/ past service
- Any legislative requirement
- Fixed or variable contributions

- Optional or Mandatory
- Contributory or non-contributory
- Tax benefits
- Death in service insurance or underpin

(ii)

In a DC scheme the benefit which will be provided to the employee [ED (finance)] is not known in advance; it will depend on:

- The contributions that are paid, which is defined as 12% of Basic salary. The actual increases in salary in the future would determine the actual contributions that would be paid
- The investment returns on the funds in the period before retirement. This would in turn depend on the timing of contributions, the choice of the pension fund manager, the type of fund, the earnings declared and the macro economic factors impacting the return &
- The cost of buying a pension at retirement from the annuity service providers. This would in turn depend on the economic scenario prevalent at the time of purchase, the number of players, the level of competitiveness in the market etc.
- We could however demonstrate the pension benefit (gross) payable assuming some scenarios on the salary escalation rate, investment return in the future and the cost of buying pension assuming no decrements up to retirement. The scenarios chosen should clearly articulate the key dependencies for the pension benefit and caveat the risks involved.

[10]**Solution 8 :****(i)**

- o **Risk**
- o Bonds (issued by governments and large corporates) have generally stable market prices than equities
- o In the event of default, bond holders generally rank higher than equity holders and have a better chance of recovering their investment amounts
- o Bonds are a good match for fixed nominal liabilities, whereas equities are a good match for real liabilities
- **Return**
- o Equities are likely to give more overall return than bonds in medium to long term
- o Equities are likely to produce real returns (i.e. higher returns when inflation is high), whereas fixed interest bonds produce nominal returns

(ii)

- **Trustees**
- o Trustees will be concerned with security of promised benefits and ongoing solvency of the sponsor and the scheme
- o Since the scheme is new, disinvestment of funds is unlikely therefore bonds can be held to maturity. In such a situation, returns from the investments are practically fixed with very little uncertainty
- o However, since the liabilities of a DB pension scheme are real, while returns from the bonds would be fixed, it is likely that in future that liabilities may grow at a faster pace than the assets, especially in a high inflation scenario, increasing the chances of insolvency in future
- **Members**
- o Members would be concerned with security of promised benefits and maximization of their own discretionary benefits

- o From the viewpoint of members, the current investment strategy can be considered safe, given the specific circumstances of the scheme
- o However, the current strategy is unlikely to produce much surplus so members could not hope for much discretionary increases if this strategy continues
- **Sponsor**
 - o Sponsor would be concerned with reducing the overall cost of the scheme and reducing uncertainty of contributions
 - o Bonds are likely to produce fixed interests, which might be lower than alternative investments
 - o Since the liabilities of the scheme are real, bonds (which provide fixed returns) are not a good match. This implies that the sponsor may have to bring in more funds to make up any deficit arising out of this mismatch

(iii)

Since the liabilities of the scheme are long-term and real, some exposure into real assets would greatly reduce the mismatch risk and increase returns. Introduction of equities and index-linked bonds would be beneficial

(iv)

The following steps need to be followed:

Model Development Time horizon: Decide on time horizon to project assets and liabilities. In the given case, since funding ratio is to be calculated for next 10 years, the time horizon needs to be at least 10 years

1. **Liability cash flow model:** Regular funding model may need to be modified include the following additional functionalities:
 1. Projecting cash flows rather than just present values
 2. Projecting liability at each future valuation date over the chosen time horizon using the chosen funding method
 3. Projecting cash flows pertaining to all options and embedded guarantees
 4. Including future new entrants
2. **Number of simulations:** Decide on the number of scenarios. In this case, since projected metrics need to be calculated for each of the next 10 valuation dates, number of simulations needed would be large, e.g. 10,000
3. **Investment model:**
 1. A stochastic investment model would need to be developed/selected; e.g. Wilkie model
 2. The model will require a number of parameters to be specified (e.g. mean, variance and probability distributions of various investment returns), links and correlations between various parameters (e.g. correlation between bond and equity returns)
4. **Valuation data:**
 1. All of data needed for regular funding valuations will be required
 2. Information regarding future new entrants to the scheme need to be included

Performing simulations

5. Decide on an initial asset mix, say 50% bonds and 50% equities
6. Run the investment and liability cash flow models together for the chosen number of simulations
7. For each of the 10,000 simulations, calculate the value of assets and liabilities over each of the next 10 valuation dates and calculate the funding ratio
8. Once all simulations are over, assess whether the chosen asset mix has met the primary objective; i.e. whether the scheme has remained solvent over next 10 years in 9,500 simulations

9. If the primary objective has been met, the investment strategy is short-listed to be considered while assessing secondary objective, else the investment strategy is rejected.
10. Repeat steps 6 through 10 by varying initial asset mix in each case.
11. List down asset allocations that have been shortlisted and rank them in order of maximum funding ratio at the end of chosen time horizon; i.e. 10th year in this case. Select the strategy that has resulted in the highest overall funding ratio at the end of 10th year.

[16]

Solution 9 :**(i)**

The assumptions relevant for consideration under the different funding methods are as below:

- the discount rate (i);
- the rate at which earnings grow (e);
- the rate at which pensions increase in payment (p);
- the rate at which leaving service benefits are revalued (r);
- the age at which members retire from the scheme (R);
- the age at which members enter the scheme (entry age); and
- the rate at which new members enter the scheme (new entrant rate).

(ii)**Standard Contribution Rate**

The contribution rate (employer and employee) appropriate to a particular funding method before taking account any Actuarial Surplus. It is normally expressed as a percentage of pensionable pay.

Modified Contribution Rate

The contribution rate (employer and employee) obtained by adjusting the Standard Contribution Rate to allow for any Actuarial Surplus. It is normally expressed as a percentage of pensionable pay. There are various ways of amortizing the Actuarial Surplus and hence of adjusting the Standard Contribution Rate. The method used should be appropriate for the purpose.

(iii)**Attained Age method**

Standard contribution rate (%)

$$= 117,000,000 / 9,000,000 = 13$$

Actuarial liability (INR)

$$= 300,000,000 + 50,000,000 + 150,000,000 = 500,000,000$$

Entry Age method

Standard contribution rate (%)

$$= 18,920 / 1\,720 = 11$$

Actuarial liability (INR)

$$= 300,000,000 + 117,000,000 + 50,000,000 + 150,000,000 - 9,000,000 \times 11.0 = 518,000,000$$

Projected Unit method

Actuarial liability (INR)

$$= 300,000,000 + 50,000,000 + 150,000,000 = 500,000,000$$

Standard contribution rate (%)

$$= 12,000,000 / 1,000,000 = 12$$

Current Unit method

Actuarial liability (INR)

$$= 240,000,000 + 50,000,000 + 150,000,000 = 440,000,000$$

Standard contribution rate – accrual bit (%)

$$= 8,000,000 / 1,000,000 = 8$$

Standard contribution rate – revaluation bit (%)
 $= (256,000,000 - 240,000,000) / 1,000,000 = 16$
 Total standard contribution rate (%) = 24

[15]**Solution 10 :****(i)**

The actual experience of a scheme should be monitored:

- To check whether the method and assumptions adopted for financing the benefits continue to be appropriate. For example, to check whether an initially assumed post-retirement mortality assumption of 100% of a certain mortality table, continues to be valid
- If not, what changes should be made in order to achieve the desired level of prudence. For example, if post-retirement mortality experience is turning out to be 90% of a certain mortality table, what revision should be made to that assumption for the purpose of triennial actuarial valuations
- To check if the scheme benefits, rules and policies are financially viable or can be sustained in future. For example if discretionary pension increases are turning out to be too expensive, due to a combined effect of increasing longevity, the practice of awarding discretionary increases may be stopped altogether

(ii)**Common considerations**

- Consideration should be given to whether the period under investigation was typical and whether the experience is likely to be representative of future experience
- Consideration would also need to be given that the period of investigation does not contain any significant abnormal events or random fluctuations that could have resulted in biased estimates of mortality
- Consideration should be given to whether analysis has been performed on sufficiently homogeneous groups that are expected to exhibit similar levels of mortality. If analysis has not been done on homogeneous groups, it would be important to ensure that the individuals to whom the investigation related are still relatively homogeneous with the individuals whose benefits will be affected by future experience

Post-retirement mortality

- It would be essential to consider and identify any trend during the analysis, e.g. mortality improvements. Consideration should be given to whether there is any reason why these trends may continue into future and assumptions about future experience should be modified accordingly
- Consideration should be given to the overall volume and credibility of data from which the results have been derived

Salary escalation

- It would be essential to consider whether the implied salary escalation rates are biased because of the impact of economic cycles. It would be essential to identify and remove the impact of such cycles before incorporating results of the analysis
- Consideration should be given to other correlated macro-economic factors over the investigation period and expected to prevail over future periods.

[8]**[Total Marks – 100]**
